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Electronic spectra and transitions of the fullerene C_{60}

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Absorption spectra of C_{60} have been measured in the ranges (a) 190–700 nm in *n*-hexane solutions at 300 K, (b) 390–700 nm in *n*-hexane and in 3-methylpentane solutions at 77 K. 40 vibronic bands were observed. They exhibit a large range of bandwidths and intensities, whose significance is discussed. Assignment of electronic transitions has been carried out using the results of theoretical calculations. Vibronic structures have been analyzed within the framework of theories of electronic transitions of polyatomic molecules applied to the I_h symmetry group. Nine allowed $^1T_{1u}$–^1A_g transitions have been assigned in the 190–410 nm region. Observed and calculated allowed transition energies and oscillator strengths are compared. Detailed vibronic analyses of the 1 $^1T_{1u}$–1 ^1A_g and 2 $^1T_{1u}$–1 ^1A_g transitions illustrate the role of Jahn–Teller couplings. Orbitally forbidden singlet–singlet transitions are observed between 410 and 620 nm. Their vibronic structures were analyzed in terms of concurrent Herzberg–Teller and Jahn–Teller vibronic interactions. The 77 K spectra provided useful information on hot bands and on other aspects of the analyses. Vibronic bands belonging to triplet←singlet transitions were detected between 620 and 700 nm.

1. Introduction

Much theoretical effort has gone into prediction of the electronic states and spectroscopy of the fullerene C_{60} [1–8]. However, the published electronic spectra have been low resolution film [9,10] or hexane solution [11–13] ultraviolet–visible absorption and have not been analyzed. The present work discusses principally room temperature hexane solution visible/UV absorption spectra of C_{60}, somewhat better resolved than earlier studies. Additional measurements were made at 77 K for solutions of C_{60} in *n*-hexane and in 3-methylpentane and for oxygen-saturated solutions at room temperature. Analysis of the spectral data has been done in terms of expectations of C_{60} electronic states and allowed and forbidden transitions. Although this is a study of spectra exhibiting solution broadened bands even at high spectral resolution, the analytical approach to the information obtained provides understanding of the electronic spectroscopy and states of C_{60} and should serve as a framework for study and analysis of high resolution gas phase electronic spectra of this fullerene when such spectra become available. In the meantime, the results of the present study should be useful in encouraging further experimental and theoretical work, including that within an astrophysical context.

2. Experimental

C_{60} samples produced and purified according to methods described elsewhere [12,14] were dissolved in *n*-hexane to form quasi-saturated solutions. The absorption spectra of this and further diluted solutions were measured in 1 and 10 cm cells, at room temperature, using a Cary 210 spectrophotometer in double beam mode. Scans were made with instrumental bandwidths of 1 nm over the whole wavelength region and also at 0.25 nm bandwidth between 390 and 700 nm. The spectra obtained in the 190–700 nm region are shown in fig. 1 and selected parts in fig. 2. Additional absorption measurements were made at $\lambda > 390$ nm, in 10 cm cells, of C_{60}-*n*-hexane solutions purged with (a) nitrogen and (b) oxygen.

Absorption spectra of saturated solutions in *n*-hex-

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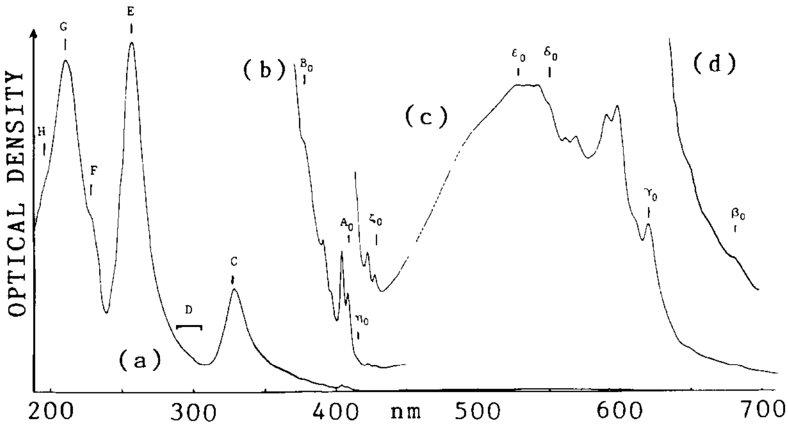

Fig. 1. Room temperature absorption spectra of C_{60}–n-hexane solutions. (a) 190–700 nm; $l=1$ cm, $R=1$ nm (b) 365–450 nm; $a\times20$. (c) 410–700 nm; $l=10$ cm, $R=1$ nm. (d) 630–700 nm; $c\times5$. Spectra measured with a Cary 210 spectrophotometer. Optical density scales differ for the four spectra.

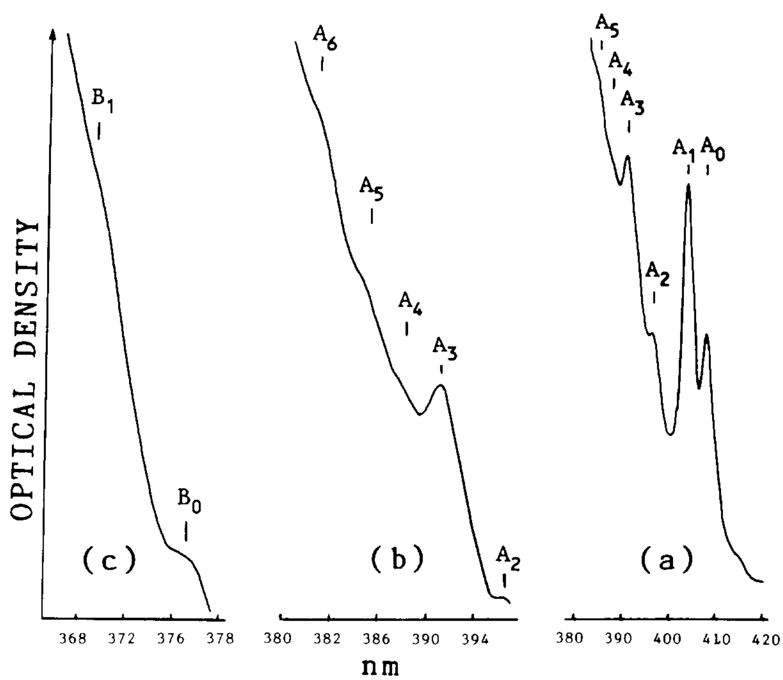

Fig. 2. Room temperature absorption spectra of C_{60}–n-hexane solutions. Details of the $\tilde{A}$ and $\tilde{B}$ system bands. (a) 380–420 nm; $l=10$ cm, $R=0.25$ nm. (b) 380–396 nm; $l=1$ cm, $R=0.25$ nm. (c) 366–380 nm; $l=1$ cm, $R=0.25$ nm. Spectra measured with a Cary 210 spectrophotometer. Optical density scales differ for the three spectra. Note change of wavelength scale in (b) and (c) with respect to (a).

ane were also measured at room temperature with a Bruker Fourier transform spectrometer (Model IFS 120 HR) in the 6000–26000 cm^{-1} (1.67 μm–385 nm) range, using 1 and 10 cm cells. Average scan numbers were 200, with some scans repeated up to 500 times. Resolution was generally 1 cm^{-1}. A silicon photodiode sensitive from the blue to ≈1.1 μm was used for some measurements; others were done with a liquid nitrogen cooled InSb detector for recordings to 1.67 μm.

The procedure with the Fourier transform spectra was to measure successively the solvent and solution transmittance; the ratio of these transmittances was then derived through a standard computer program. Appropriate optical filters were used in the various wavelength regions. Recording times were about 20 min for specific segments of the spectra. Spectra obtained as transmittances over the spectral range 400–700 nm are shown in fig. 3 where they are compared with the absorbance mode spectra measured with the Cary spectrophotometer.

The FTR spectrometer was also employed in two other types of absorption measurements: (i) low temperature (77 K) spectra of (a) polycrystalline solutions of C_{60}-n-hexane (9000–20000 cm^{-1}), cell thickness ≈3 mm, and (b) transparent glassy-matrix solutions of C_{60}-3-methylpentane (9000–26000 cm^{-1}), cell thickness 6 mm; (ii) room temperature spectra of C_{60}-n-hexane solutions purged by shaking with (a) nitrogen, (b) oxygen, in 10 cm cells.

3. Results

A number of features existing as shoulders or otherwise insufficiently resolved in earlier published spectra are much more clearly evident in the spectra we have measured. In addition, the use of higher resolution than in previous studies enabled us to obtain more precise wavelengths (frequencies) for observed features. The Fourier transform spectra allowed us to extend previous observations and to discover very weak bands at $\lambda > 620$ nm which were then also recorded with the Cary spectrophotometer (fig. 3). Peak profiles are identical for the spectra recorded with the Cary spectrophotometer at $R=0.25$ nm (≈16 cm^{-1}) and with the Bruker Fourier transform spectrometer at $R=1$ cm^{-1}.

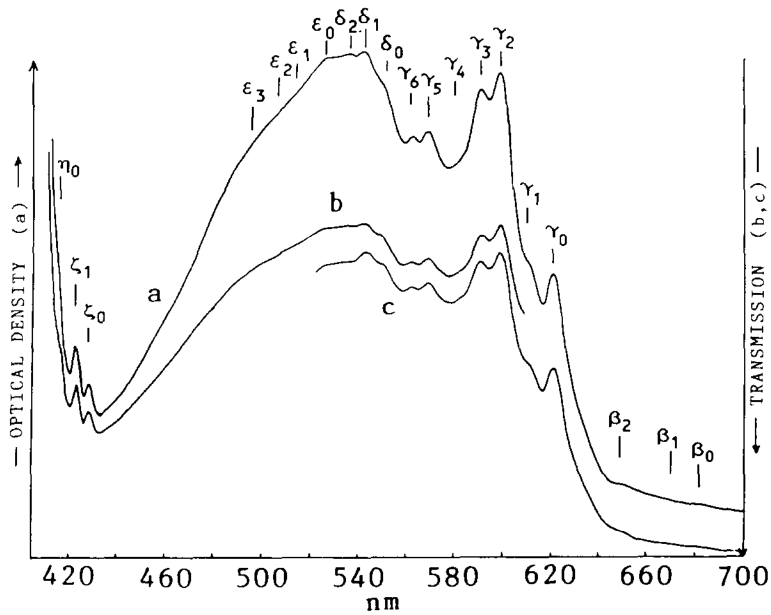

Fig. 3. Room temperature absorption spectra of C_{60}–n-hexane solutions. (a) 400–700 nm. Absorbance measured with a Cary 210 spectrophotometer, $l=10$ cm, $R=0.25$ nm. (b) 400–700 nm. Transmittance measured with a Bruker IFS 120 HR Fourier transform spectrometer, $l=10$ cm, $R=1$ cm^{-1}. (c) 520–700 nm as for (b).

Table 1
Absorption spectrum of C_{60} at 399 K; band wavelengths, frequencies and assignments

Band code	λ (nm)	ν (cm^{-1})	Assignment
α [a]	$\geqslant$695	$\leqslant$14390	$1\,^3T_{2g}$–$1\,^1A_g$; $0^0_0+t_{1u}$, t_{2u}, g_u or h_u
β_0	682.1	14661	$1\,^3T_{1g}$–$1\,^1A_g$; $0^0_0+a_u$, t_{1u}, t_{2u}, g_u or h_u
β_1	670.8	14907	$\beta_0+h_g(8)$
β_2	648.4	15423	$\beta_0+h_g(5)$
β_3 [b]	627.5	15935	$\beta_0+h_g(3)$
γ_0	620.2	16124	$1\,^1T_{1g}$–$1\,^1A_g$; $0^0_0+a_u$, t_{1u} or h_u
γ_1	610.1	16390	$\gamma_0+h_g(8)$
γ_2	598.1	16720	interleaved vibronic bands
γ_3	590.3	16941	of $1\,^1T_{2g}$–$1\,^1A_g$ and
γ_4	580.0	17242	$1\,^1T_{1g}$–$1\,^1A_g$
γ_5	568.0	17600	(see text)
γ_6	561.9	17798	
δ_0	549.3	18205	$1\,^1T_{2u}$–^1A_g; $0^0_0+g_g$ or h_g
δ_1	542.6	18430	$\delta_0+h_g(8)$
δ_2	534.2	18718	$\delta_0+a_g(2)$
ε_0	526.0	19011	$1\,^1H_u$–$1\,^1A_g$; $0^0_0+t_{1g}$, t_{2g}, g_g or h_g
ε_1	509.9	19612	$\varepsilon_0+a_g(2)$
ε_2	502.2	19911	$\varepsilon_0+g_g(?)$
ε_3	496.0	20161	$\varepsilon_0+h_g(3)$
ζ_0	427.9	23372	$2\,^1H_u$–$1\,^1A_g$; $0^0_0+t_{1g}$, t_{2g}, g_g or h_g
ζ_1	422.5	23669	$\zeta_0+h_g(8)$
η_0	415.0	24098	$2\,^1G_u$–$1\,^1A_g$; $0^0_0+t_{2g}$, g_g or h_g
η_1 [b]	410.8	24345	$\eta_0+h_g(8)$
A_0	408.3	24490	$1\,^1T_{1u}$–$1\,^1A_g$; 0^0_0
A_1	404.0	24754	$A_0+h_g(8)$
A_2	396.5	25220	$A_0+h_g(8)+a_g(2)$
A_3	391.0	25575	$A_0+h_g(4)$
A_4	387.8 sh [c]	25786	$A_0+h_g(3)$
A_5	385.0 sh	25974	$A_0+a_g(1)$
A_6	380.8 sh	26261	$A_0+h_g(8)+a_g(1)$
B_0	377.0 sh	26525	$2\,^1T_{1u}$–$1\,^1A_g$; 0^0_0
B_1	369.4 sh	27071	$B_0+a_g(2)$
B_2	367.2 sh	27233	$B_0+h_g(6)$
B_3	358.4 sh	27871	$B_0+a_g(1)$
C	328.4	30451	$3\,^1T_{1u}$–$1\,^1A_g$
D	305–285	32786	$4\,^1T_{1u}$–$1\,^1A_g$
		35088	$5\,^1T_{1u}$–$1\,^1A_g$
E	256.6	38971	$6\,^1T_{1u}$–$1\,^1A_g$
F	227.4	43975	$7\,^1T_{1u}$–$1\,^1A_g$
G	211.0	47393	$8\,^1T_{1u}$–$1\,^1A_g$
H	195	51282	$9\,^1T_{1u}$–$1\,^1A_g$

[a] O_2 induced absorption. [b] Observed at 77 K, see text. [c] sh = shoulder or inflexion

Table 1 lists the measured wavelengths, frequencies, and assignments of the observed features in room temperature solution spectra. Precision in peak frequency measurements is estimated to be ± 5 cm^{-1} for sharp peaks, ± 10 cm^{-1} for broad peaks, ± 20 cm^{-1} for shoulders and ± 50 cm^{-1} for inflexions. Extinction coefficients were obtained from optical densities which were scaled to the $\epsilon_{max} = 175000$ value at

$\lambda = 256$ nm reported by Hare et al. [12]. They are in reasonable agreement with previously published [12,13] values. For example in the strong transition region, our results, compared with those of Allemand et al. [13] (in parentheses) are as follows: λ (nm), $\log \epsilon_{max}$: 211 nm, 5.20(5.17); 227 nm, 4.95(4.91); 256 nm, 5.24(5.24); 328 nm, 4.72(4.71); 377 nm, 3.79(3.75); 404 nm, 3.47(3.48); 408 nm, 3.27 (3.28).

We consider four spectral regions in the set of bands listed in table 1.

(1) The strong band region between 190 and ≈ 350 nm which has 3 intense broad peaks at 211, 256 and 328 nm; in addition there are shoulders or inflexions at ≈ 195, 227 and 295 nm (fig. 1).

(2) A region of much weaker bands between 350 and 430 nm some of which are sharply structured (figs. 1 and 2) and appear to be due to electronic transitions exhibiting some vibrational structure.

(3) A broad weak continuum between 430 and 640 nm, whose maximum is at about 540 nm, and on which are superposed several clearly evident peaks and some weaker shoulders or inflexions (figs. 1 and 3). These bands also appear to be due to vibronic transitions.

(4) Three extremely weak but definite bands between 640 and 690 nm (fig. 3).

The low temperature spectra showed only minor changes with respect to the 300 K spectra. The bands showed diminished half-widths at 77 K; a few bands hidden at 300 K were revealed in the spectra more sharply resolved at 77 K. Interesting intensity modifications were observed in comparing the N_2 and O_2 "saturated" room temperature spectra. The significance of these observations will be presented below.

4. Discussion

We will discuss the four spectral regions in turn. We consider the principal bands in the 190–410 nm region to be mainly due to orbitally allowed singlet–singlet transitions. The features in the region at wavelengths greater than 410 nm we assign exclusively to forbidden transitions which appear because of Herzberg–Teller vibronic interactions and also to some spin-forbidden triplet$\leftarrow$singlet transitions. Understanding the various transitions requires first a discussion on theoretical expectations of the electronic spectra of C_{60}.

4.1. Calculated electronic states and transitions of C_{60}

The C_{60} molecule contains 60 identical carbon atoms arranged in 20 six-membered and 12 five-membered rings; each atom lies at the vertex of one five-membered and two six-membered rings. Thus C_{60} is a truncated isocahedron belonging to the I_h symmetry group. Its carbon–carbon bonds are of two types; 30 equal bonds are "short", of length variously estimated to be between 1.370 and 1.411 Å, and 60 are equal "long" bonds of between 1.439 and 1.474 Å in length [1–8,15].

The totally symmetric ^1A_g ground state of C_{60} issues from the closed shell electron configuration: ... $a_g^2 t_{1u}^6 h_g^{10} t_{2u}^6 g_u^8 g_g^8 h_g^{10} h_u^{10}$. For this non-alternant species, the first two excited configurations, and corresponding state symmetries are:

... $h_g^{10} h_u^9 t_{1u}$ (i.e. the LUMO configuration): T_{1g}, T_{2g}, G_g, H_g
... $h_g^{10} h_u^9 t_{1g}$: T_{1u}, T_{2u}, G_u, H_u

In addition there is a (HOMO-1)$\rightarrow$(LUMO) excitation possible, where the excited configuration is $h_g^9 h_u^{10} t_{1u}$, giving states of the same symmetry as the ... $h_g^{10} h_u^9 t_{1g}$ configuration.

Calculations on the electronic states of C_{60} have been carried out by a variety of methods [1–8] but often only allowed transitions are reported. We are principally concerned here with those studies which give energies not only of orbitally allowed transition excited states, but also states involved in orbitally forbidden and spin forbidden transitions [3,5,6]. We discuss first of all the singlet excited states which are the principal states that give rise to discernable absorption strength.

Laszlo and Udvardi [6] carried out PPP CI molecular orbital calculations and found 36 singlet excited electronic states in the energy range 2.6–7.6 eV of which 24 lie below 6.2 eV; the QCFF/PI method calculations of Negri et al. [5] lead to 16 states between 2.6 and 6.2 eV. Triplet states were also calculated by these two groups [3,5,6]. CNDO/S calculations involving very large numbers of electronic configurations have been done by Braga at al. [8]. The results are incompletely published in that information is given only on the allowed singlet transition states; nevertheless they provide much useful information for our analysis.

4.2. The allowed transitions

4.2.1 Allowed transition assignments

Since C_{60} belongs to the I_h symmetry group, the only transitions between the ^1A_g ground state that are associated with an electric dipole moment must have $^1T_{1u}$ ($^1F_{1u}$ in an alternative notation [16]) upper electronic states. The oscillator strengths calculated by Braga et al. [8] for allowed $^1T_{1u}$–$^1A_{1g}$ transitions are given in table 2. It should be noted that the calculated oscillator strength of 1 $^1T_{1u}$–1 ^1A_g, the weakest of the allowed transitions, depends markedly on the carbon–carbon bond lengths and on whether equal or alternating bond lengths are used [4,7]. For example, this value varies from 0.07 to 0.27 for INDO/S calculations using various sets of bond length [7].

The calculation method and size of the configuration set also affect the oscillator strengths of individual $^1T_{1u}$–^1A_g transitions as evidenced by a comparison between the results of Laszlo and Udvardi [3,6] (134 configurations), Negri et al. [5] (196 configurations) and Braga et al. [8] (up to 900 configurations). In these calculations only singly excited configurations were considered.

We have chosen in the first place to relate our experimental findings to the results of the calculations of Braga et al. [8]. Their method treats the σ–π interaction, which must be taken into account in a non-planar system such as C_{60}, in more satisfactory fashion than the PPP method of Laszlo and Udvardi [3,6] or the QCFF/PI technique of Negri et al [5]. Furthermore, a much larger number of configurations was employed in the CNDO/S calculations of Braga et al. [8].

CNDO/S calculations involving 808 and 900 configurations predict that about 12 allowed $^1T_{1u}$–^1A_g transitions should exist between 3.4 and 6.9 eV (fig. 2 of ref. [8]). The first allowed transition is calculated to be at 3.4 eV with an oscillator strength $f=0.08$. The three most intense $^1T_{1u}$–^1A_g transitions are predicted to lie at 4.38, 5.24 and 5.78 eV. The calculated oscillator strengths given in table 2 were obtained with a 808 configuration calculation; only small differences, <15%, in transition intensity values occur on going to 900 configurations.

We have assigned 9 allowed $^1T_{1u}$–^1A_g transitions in the 190–410 nm (3–6.5 eV) region (table 2). The *relative areas* of the main peaks C, E and G agree well with those predicted from the calculated f values of Braga et al. [8]. Agreement is less good for the weaker features D and F; the corresponding peak areas are more uncertain and, furthermore, the calculated f values for these weaker allowed transitions should be more sensitive than the strong peaks to the quality of the configuration interactions.

Approximate *oscillator strengths* determined from the observed spectra are also given in table 2. For the structureless bands these were obtained using the simplified expression [17] linking the transition oscillator strength f to the frequency half-width ($\Delta\nu_{1/2}$

Table 2
$^1T_{1u}$–1 ^1A_g allowed transitions: observed and calculated energies and oscillator strengths

Excited state	E_{obs} (eV)	E_{calc} (eV) [8]	E_{obs}/E_{calc}	f_{obs}	f_{calc} [8]	f_{obs}/f_{calc}
1 $^1T_{1u}$	3.04	3.4	0.89	0.015± 0.005	0.08	0.19
2 $^1T_{1u}$	3.30	4.06	0.81		0.41	
3 $^1T_{1u}$	3.78	4.38	0.86	0.37	2.37	0.16
4 $^1T_{1u}$ }	{ 4.06	4.70	(0.86) }	0.10 }	0.3 }	0.33
5 $^1T_{1u}$ }	{ 4.35	5.07	(0.86) }			
6 $^1T_{1u}$	4.84	5.24	0.92	2.27	7.88	0.29
7 $^1T_{1u}$	5.46	5.54	0.99	0.22	1.18	0.19
8 $^1T_{1u}$	5.88	5.78	1.02	3.09	10.74	0.29
9 $^1T_{1u}$	6.36	6.28	1.01			

in cm^{-1}) of an assumed Gaussian-shaped band and its peak extinction coefficient ϵ_{max}:

$$f=4.32\times10^{-9}\,\Delta\nu_{1/2}\epsilon_{max}(\pi/\ln 2)^{1/2}.$$

The influence of the solvent (polarizability) is ignored in this expression. Gas phase f values of C_{60} can be estimated, from the refractive index of n-hexane, to be up 20% smaller than the solution values. Estimates of the oscillator strengths of some of the vibrationally structured transitions were obtained from peak areas normalized to those of the more intense unstructured bands.

The relative energies and intensities of the observed absorption peaks are in good agreement with the calculated values for the allowed transitions, as illustrated by the ratios of observed to calculated values in table 2. The agreement between the absolute observed and calculated energies is satisfactory. The calculated f values are 3–5 times greater than the observed values. However, the absolute values of oscillator strengths of large polyatomic molecules are notoriously difficult to calculate within a factor of this order of magnitude unless highly excited configurations are included in the calculations. In this connection we note that only singly excited configurations were used by Braga et al. [8]. Inclusion of doubly excited configurations would be expected to diminish the calculated f values. Indeed it has been found for benzene and for naphthalene that inclusion of doubly excited configurations hardly modifies calculated state energies but reduces calculated oscillator strength by factors of the order of 2 to 4 [8], i.e. the same order of magnitude as f_{calc}/f_{obs} in our C_{60} studies [#1].

4.2.2. Bandwidths

The invariance of the band profiles with spectral resolution ($R\approx 16\ cm^{-1}$ and $R=1\ cm^{-1}$) shows that the band shapes and widths are intrinsic to the solutions.

The broadness of the strong bands in the 190–350 nm region merits some comment. Broad intense absorption bands with little or no vibrational structure are often observed in solution spectra. For example, strong transitions of large aromatic compounds in room temperature spectra in paraffinic solvents invariably give rise to broad absorption bands having a single peak and showing very little trace of vibrational structure [18].

One of the important contributing factors can be strong solute–solvent interaction for intense transitions. The breadth of the band depends on the electronic transition densities in the interacting molecules [19]. The transition densities in these molecules will be greatest for the transitions of highest oscillator strength. The f values for the C_{60} solute are given in table 2; the paraffinic solvents begin to absorb at $\lambda<200$ nm. Thus the breadth of the electronic bands of C_{60} in n-hexane solution can be expected to be large for the higher energy transitions. Among the allowed transitions, only those of smallest oscillator strength, $1(2)\ {}^1T_{1u}$–$1\ {}^1A_g$, exhibit vibrational structure in solution. The forbidden transitions at $\lambda>408$ nm also show many vibrational features. Thus the bandwidths and structures observed over the whole spectral region investigated are consistent with the solute–solvent interaction parameters under discussion.

Other possible factors contributing to absorption band breadth are vibronic state congestion and hot bands for species with very many vibrational modes. Our low temperature studies relate to these questions, as discussed later.

Another possible factor concerns important distortions of symmetry coordinates which can occur on excitation, giving rise to large Franck–Condon factors for displaced oscillators. We note that in a set of allowed transitions, electron displacements will be greatest for those of highest oscillator strengths, so that bonding changes could be greatest for these cases. The Franck-Condon factors for the resulting displaced oscillators should therefore be large. However, because of the large number of (approximate [1]) π electrons in C_{60}, it is unclear, without vibronically resolved gas phase spectra, or specific calculations, how displaced the oscillators will be on electron excitation. Calculations have been carried out [5] only for the lowest, which is also the weakest, of the allowed ${}^1T_{1u}$–${}^1A_g$ transitions. We will return to this aspect in discussion of the vibrational structure of the observed weak transitions.

We will discuss in more detail the first allowed

[#1] Note added in proof: Improved CI calculations with 1174 configurations (ref. [39]) do not modify C_{60} state energies, but yield oscillator strengths much closer to observed values, $f_{obs}/f_{calc}=0.4$ to 0.9, as compared with 0.19 to 0.33 in table 2.

transition, 1 $^1T_{1u}$–1 ^1A_g whose origin we assign to the 408.3 nm band, and the 2 $^1T_{1u}$–1 ^1A_g transition whose origin band is at 377 nm.

4.2.3. The 1 $^1T_{1u}$–1 ^1A_g transition

Assignment of the 408.3 nm band system ($\tilde{A}$ system) to the first allowed 1 $^1T_{1u}$–1 ^1A_g transition is supported by the reasonably good agreement between relative calculated and observed energies and transition strengths (table 2).

We remark, furthermore, the good agreement between the observed optical transition energy (3.04 eV) and the interval ($\approx$3eV) between the first and third photoelectron bands in the ultraviolet photoelectron spectrum (UPS) of the anion C_{60}^- [20]. It is in this energy region that the bulk of the lowest lying ungerade states should occur since, on a single configuration basis, the first three C_{60}^- photoelectron bands correspond to the following electron removals to form either gerade or ungerade Koopmans states:

1st PES band: $...h_g^{10}h_u^{10}t_{1u} \to ...h_g^{10}h_u^{10}$ (1 ^1A_g)
2nd PES band: $...h_g^{10}h_u^{10}t_{1u} \to ...h_g^{10}h_u^{9}t_{1u}$
(gerade electronic states)
3rd PES band: $...h_g^{10}h_u^{10}t_{1u} \to ...h_g^{9}h_u^{10}t_{1u}$
(ungerade electronic states)

This interpretation is consistent with the results of MO calculations of C_{60} states [5,6] (fig. 4).

We now discuss details of the absorption spectrum and vibronic assignments corresponding to the 1 $^1T_{1u}$–1 ^1A_g transition shown in fig. 2. We note first of all that the triply degenerate 1 $^1T_{1u}$ state should exhibit Jahn–Teller effects (see section 4.3).

The 0_0^0 band (A_0) is followed by the A_1 band 264 cm^{-1} to higher frequencies which is assigned to the Jahn–Teller active vibronic transition involving the lowest frequency h_g vibration (table 3), i.e. $h_g(8)$ [21,22]. In the ground state this mode has a frequency 273 cm^{-1} and corresponds to a squashing vibration. The radial distortion would tend to deform the spherical C_{60} fullerene towards an ellipsoidal structure. Important activity of this vibrational mode in the 1 $^1T_{1u}$–1 ^1A_g transition was not expected from the mode displacement calculations of Negri et al. [5] and points to insufficiencies in their description of the 1 $^1T_{1u}$ state and possibily also in their estimation of the change of equilibrium coordinates due to excitation. The C_{60} ground state is probably well described, since the vibrational frequencies calculated by Negri et al. [5] are closer to the observed Raman and infrared frequencies [19,20] than those calculated by other authors [23–27].

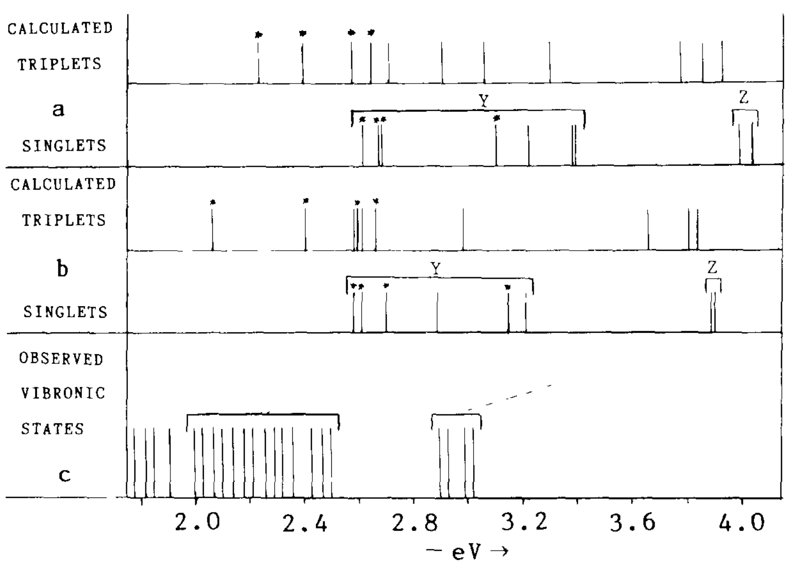

Fig. 4. Energies of C_{60} electronic states below the first allowed singlet 1 $^1T_{1u}$. (a) Triplets and singlets calculated by the 3D-PPP-CI method [6]. Asterisks indicate gerade states. (b) Triplets and singlets calculated by the QCFF/PI method [5]. Asterisks indicate gerade states. (c) Observed vibronic states.

The A_2 band at 25220 cm^{-1} is assigned as a combination between one quantum of the $h_g(8)$ Jahn–Teller inducing mode ($\nu = 264$ cm^{-1}) and one quantum of the in-phase ring breathing vibration $a_g(2)$. The 1 $^1T_{1u}$ state value of the latter would therefore be 466 cm^{-1}, as compared to the ground state value 497 cm^{-1} observed in Raman spectra [22]. This implies that the corresponding oscillator is only a little displaced during the 1 $^1T_{1u}$–1 ^1A_g transition in agreement with the small displacement parameter calculated by Negri et al. [5] for this breathing mode. An alternative possible assignment for A_2 is as $0_0^0 + h_g(6)$ since the interval A_2–A_0 = 730 cm^{-1} is close to the ground state frequency $h_g(6) = 710$ cm^{-1}.

The feature A_3 at +1085 cm^{-1} from the 0_0^0 band can be assigned to $0_0^0 + h_g(4)$. This Jahn–Teller inducing mode was also predicted to be little active in the 1 $^1T_{1u}$–1 ^1A_g transition. The ground state calculated frequency for the Jahn–Teller inducing vibration $h_g(4)$ is about 1155 cm^{-1} [5,24]; the observed Raman band at 1099 cm^{-1} [22] probably corresponds to this mode. Likewise, assignment of A_4 as $0_0^0 + h_g(3)$ gives $h_g(3) = 1296$ cm^{-1}, close to an observed ground state Raman frequency 1250 cm^{-1}.

The A_5 band at 0_0^0 + 1484 cm^{-1} is assigned to $0_0^0 + a_g(1)$. Its upper state value, 1484 cm^{-1} is close to the observed ground state value 1470 cm^{-1} [23]. Negri et al. [5] predict that the totally symmetric out-of-phase breathing vibrational mode $a_g(1)$ should be strongly active in the 1 $^1T_{1u}$–1 ^1A_g transition. The A_6 band corresponds to the combination $h_g(8) + a_g(1)$.

Our band assignments are thus in qualitative agreement with Negri et al. calculations of displacement parameters in 1 $^1T_{1u}$–1 ^1A_g for the two a_g vibrations. Agreement is less good for the h_g modes. This is most probably due to an insufficient number of excited electronic configurations and in particular to the absence of multi-excited configurations in the description of the $^1T_{1u}$ state, a defect which also shows up in the far too large oscillator strength calculated for the 1 $^1T_{1u}$–1 ^1A_g transition by Negri et al. [5].

We note that a_g and h_g modes that appear in the 1 $^1T_{1u}$–1 ^1A_g transition are also active in the vibrationally induced forbidden transitions which we address later in a single mode approximation.

4.2.4. The 2 $^1T_{1u}$–1 ^1A_g transition

The $\tilde{B}$ system consists of 4 shoulders, between 358 and 377 nm, on the low frequency flank of the strong 328 nm band (figs. 1 and 2). Other inflexions may possibly exist in the 347–356 nm region. We have assigned $\tilde{B}$ to the 2 $^1T_{1u}$–1 ^1A_g allowed transition. There is good agreement with the calculated energy of the 2 $^1T_{1u}$ state (table 2). Assignment of the vibronic bands of this system gives the following vibrational frequencies for the 2 $^1T_{1u}$ state (ground state frequencies in parentheses): $a_g(2) = 546(497)$ cm^{-1}, $h_g(6) = 708(710)$ cm^{-1}, $a_g(1) = 1346(1469)$ cm^{-1}.

4.3. Orbitally forbidden electronic transitions

All singlet excited electronic states other than those of $^1T_{1u}$ symmetry correspond to transitions forbidden in absorption from the 1 ^1A_g ground state for the isolated molecule.

Orbitally forbidden bands can appear weakly in solution spectra due to reduction of local symmetry resulting from solute–solvent interactions. This is unlikely to be important for a spherical molecule the size of C_{60}, whose diameter can be estimated to be about 7.1 Å, since the intermolecular forces acting on the solute would tend to be isotropic, even on a short time scale.

Orbital forbiddenness can be partially lifted for the isolated molecule by Herzberg–Teller interactions in which excitation of a vibration of suitable symmetry enables the forbidden transition to "steal" intensity from allowed transitions. In the case of C_{60}, the false origins thus created should be capable of acting as origin bands for excitation not only of totally symmetric modes but also of non-totally symmetric Jahn–Teller active vibrations. We note that, apart from the (very few) A_g and A_u states, the electronic states of C_{60} are triply and higher-order orbitally degenerate. These states should exhibit Jahn–Teller distortions in which the degeneracy is removed by excitation of vibrations of suitable symmetries. For the allowed transitions $^1T_{1u}$–^1A_g, the Jahn–Teller active vibrations must be of h_g symmetry [16]. There are 8 modes in this symmetry class.

I_h symmetry group product tables [16] were used to establish the Herzberg–Teller and Jahn–Teller active modes for different types of transition (table 3). There are several vibrational modes within each Herzberg–Teller active symmetry class. For example, there are six g_g, six g_u, seven h_u and eight h_g vibra-

Table 3
C_{60} electronic transitions between the 1 ^1A_g ground state and excited electronic states. Herzberg–Teller and Jahn–Teller active modes

Excited state electronic symmetry	Spin–orbit [b] operator symmetry	Herzberg–Teller active vibrational mode symmetry	Jahn–Teller active vibrational mode symmetry
^1A_u	–	t_{1g}	–
^3A_u	T_{1g}	– [a]	–
$^1T_{1u}$	–	(allowed transition)	h_g
$^3T_{1u}$	T_{1g}	t_{1g}, t_{2g}, g_g, h_g	h_g
$^1T_{2u}$	–	g_g, h_g	h_g
$^3T_{2u}$	T_{1g}	t_{1g}, t_{2g}, g_g, h_g	h_g
^1G_u	–	t_{2g}, g_g, h_g	g_g, h_g
^3G_u	T_{1g}	t_{1g}, t_{2g}, g_g, h_g	g_g, h_g
^1H_u	–	t_{1g}, t_{2g}, g_g, h_g	g_g, h_g
^3H_u	T_{1g}	(t_{1g}, t_{2g}, g_g, h_g) [a]	g_g, h_g
^1A_g	–	t_{1u}	–
^3A_g	T_{1g}	a_u, t_{1u}, h_u	–
$^1T_{1g}$	–	a_u, t_{1u}, h_u	h_g
$^3T_{1g}$	T_{1g}	a_u, t_{1u}, t_{2u}, g_u, h_u	h_g
$^1T_{2g}$	–	g_u, h_u	h_g
$^3T_{2g}$	T_{1g}	t_{1u}, t_{2u}, g_u, h_u	h_g
^1G_g	–	t_{2u}, g_u, h_u	g_g, h_g
^3G_g	T_{1g}	t_{1u}, t_{2u}, g_u, h_u	g_g, h_g
^1H_g	–	t_{1u}, t_{2u}, g_u, h_u	g_g, h_g
^3H_g	T_{1g}	a_u, t_{1u}, t_{2u}, g_u, h_u	g_g, h_g

[a] The forbiddenness of ^3A_u–^1A_g transitions is lifted by the spin–orbit interaction alone. This is also the case for ^3H_u–^1A_g transitions some of whose spin-electronic components can have the Herzberg–Teller terms indicated in parentheses.
[b] In electron configuration space.

tional modes in C_{60}, covering a wide range of frequencies [21–27]. Thus, in principle, there could be several false origins corresponding to a particular (Herzberg–Teller active) vibrational symmetry.

In our vibronic analysis we have made the simplifying assumption that one particular Herzberg–Teller vibrational mode will be dominant in a specific forbidden electronic transition. This is indeed usually the case as is observed in the Herzberg–Teller components of several electronic transitions [28].

Before discussing vibrational structure in the forbidden transitions we first consider their electronic state assignments.

4.3.1. Forbidden transition assignments

We assign the extremely weak $\tilde{\beta}$ system bands in the 14661–15365 cm^{-1} (640–690 nm; 1.82–1.91 eV) region (figs. 1 and 3) to triplet←singlet vibronic transitions. These will be discussed later.

The bands of the $\tilde{\gamma}$, $\tilde{\delta}$, ..., $\tilde{\eta}$ systems in the 16124–24345 cm^{-1} (2.0–3.02 eV) region are assigned to forbidden singlet–singlet transitions. These bands fall into two distinct groups, with an energy gap between them: 14 bands between 2.0 and 2.5 eV; 4 bands between 2.90 and 3.02 eV. Our arguments for assigning these 18 bands to (vibronic components of) forbidden singlet–singlet transitions are as follows.

(1) The extinction coefficients of the $\tilde{\gamma}$, $\tilde{\delta}$, ..., $\tilde{\eta}$ bands are about 2 orders of magnitude greater than for the group of $\tilde{\beta}$ bands assigned to triplet←singlet transitions, but are considerably smaller than ϵ_{max} of the bands at $\lambda < 410$ nm assigned to allowed $^1T_{1u}$–^1A_g transitions.

(2) The energy range of the first group of 14 bands is similar to the 1.5–2.5 eV interval (peak-to-peak 1.9 eV) between the first and second UPS bands of C_{60}^-, whereas the second group of 4 bands corresponds in energy to the peak-to-peak interval, 3 eV, between the first and third UPS bands of C_{60}^- [20].

(3) Two groups of forbidden singlet–singlet bands

(Y and Z, fig. 4), separated by an energy gap are expected in the low energy range from the calculations of Laszlo and Udvardi [3,6] and Negri et al. [5] (the corresponding data from the CNDO/S calculations of Braga et al. [8] are not yet available). The observed bands contain vibrational features, so their density in fig. 4 exceeds that of the calculated purely electronic transitions.

Fig. 4 shows that the first group of observed vibronic bands between 2 and 2.5 eV (0.5 eV total span) is followed by an energy gap of 0.4 eV. This is matched by a span of 0.63 eV for the first group of states (Y) calculated by Negri et al. [5] followed by a gap (Y–Z) of 0.68 eV; the corresponding values for Laszlo and Udvardi's calculations [6] are 0.78 eV (span) and 0.59 eV (gap).

The two electronic states calculated to occur in the Z group of states are the 2 ^1H_u and 2 ^1G_u levels. We can thus with some confidence assign the $\tilde{\zeta}$ and $\tilde{\eta}$ bands in the 2.90–3.02 eV region, to the 2 $^1H_u \leftarrow 1\ ^1A_g$ and 2 ^1G_u–1 ^1A_g transitions. In table 1 we have followed the calculated order of these two states. On symmetry grounds (table 3) the Jahn–Teller inducing vibration could be h_g or g_g for these two transitions. We have assigned this to $h_g(8)$ on the basis of the observed band intervals (297 and 247 cm^{-1}, respectively). The g_g vibration of lowest frequency in the ground state is expected to have a value of about 500 cm^{-1} [5].

Specific assignments of the singlet–singlet transitions in the 2–2.5 eV region ($\tilde{\gamma}$, $\tilde{\delta}$, $\tilde{\varepsilon}$ systems) are more delicate. First of all, one or two pairs of transitions could be energy degenerate [3,5,6] within the 0.02 eV average half width of our room temperature bands. Thus the vibronic bands in the 2–2.5 eV region could represent a set of 7 transitions of which only a reduced number are separately resolved.

On the basis of electronic state calculations (fig. 4) we can reasonably assume that the first 3 or 4 singlet–singlet transitions are to gerade upper states. We therefore assign the $\tilde{\gamma}$ band region to vibronic features of the first two forbidden electronic transitions 1 $^1T_{2g}$–1 ^1A_g and 1 $^1T_{1g}$–1 ^1A_g, which are calculated to be quasi-degenerate [3,5,6]. A very recent magnetic circular dichroism study of C_{60} [29] indicates that the lowest energy transition is 1 $^1T_{1g}$–1 ^1A_g (table 1).

The aspect of the spectra in the $\tilde{\delta}$ and $\tilde{\varepsilon}$ band regions has led us to consider that other transitions are involved. The transitions expected from calculations are 1 ^1G_g–1 ^1A_g, $^1T_{2u}$–1 ^1A_g, 1 ^1H_u–1 ^1A_g, 1 ^1H_g–1 ^1A_g and 1 ^1G_u–1 ^1A_g. In the absence of vibronic coupling calculations we cannot say a priori what to expect for the relative intensities of such Herzberg–Teller induced transitions. In table 1, we have selected to assign 1 $^1T_{2u}$–1 ^1A_g and 1 ^1H_u–1 ^1A_g on the basis of state energy calculations. The other three transitions may also be active and may overlap to give rise to the quasi-continuum underlying the $\tilde{\delta}$ and $\tilde{\varepsilon}$ systems.

4.3.2. Vibronic structure of forbidden singlet–singlet transitions

Forbidden singlet–singlet transitions can be Herzberg–Teller induced and their upper electronic states subject to Jahn–Teller dynamic distortions. The expected vibronic structure is therefore as follows for an absorption transition.

The origin band is expected to be absent or, if solvent induced, extremely weak as compared with other vibronic bands of the same transition. The initial band of a transition at 300 K should be a false origin involving excitation of a Herzberg–Teller active mode (table 3) either in the upper or the lower state. If a lower state vibration is active, the corresponding hot band should be temperature sensitive, as discussed later. We recall the assumption that one Herzberg–Teller inducing mode will be dominant in each forbidden transition of C_{60}. Vibronic transitions based on the multiple other false origins are expected to contribute to the quasi-continuous background underlying the forbidden transitions, e.g. in the 430–640 nm region.

For each forbidden transition, there should be bands corresponding to excitation (from the false origin) of Jahn–Teller active modes, as well as of totally symmetric a_g modes, in the upper state. Evidence for Jahn–Teller h_g mode excitation in the forbidden transitions is the existence of intervals of the order of 260±40 cm^{-1}, e.g. β_1–β_0=246 cm^{-1}, γ_1–γ_0=266 cm^{-1}, δ_1–δ_0=225 cm^{-1}, ζ_1–ζ_0=297 cm^{-1}, η_1–η_0=247 cm^{-1}. These values are similar to the frequency of mode $h_g(8)$=273 cm^{-1} in the ground state and 266 cm^{-1} in the 1 $^1T_{1u}$ state. Some other intervals can perhaps be associated with the breathing vibrations $a_g(2)$ and $a_g(1)$, e.g. γ_3–γ_1=551 cm^{-1}, δ_2–δ_0=513 cm^{-1}, ε_1–ε_0=601 cm^{-1} (cf. $a_g(2)$=497 cm^{-1} in the ground state, 466 cm^{-1} in 1 $^1T_{1u}$), γ_5–

$\gamma_0 = 1476\ cm^{-1}$ (cf. $a_g(1) = 1469\ cm^{-1}$ in the ground state, $1484\ cm^{-1}$ in $1\ ^1T_{1u}$). In the case of the $1\ ^1H_u$–$1\ ^1A_g$ transition (ε band system), some of the observed intervals could be due to g_g Jahn–Teller modes.

4.4. Low temperature spectra and hot bands

Only 14 vibrational frequencies of ground state C_{60} are known, out of 46 expected (C_{60} has 174 vibrations but many are degenerate). Calculations [5] lead us to expect that mode $h_g(8)$ will have the smallest frequency, with the next highest about 100 cm^{-1} greater in value. Thus $h_g(8)$, whose frequency is 273 cm^{-1} in the ground state, should be the most important hot band starting level, at least from the viewpoint of relative vibrational level populations at 300 K.

The direct products of irreducible representations of the I_h point group allow a transition from a h_g mode hot ground state level to all of the Herzberg–Teller induced vibronic levels of all possible electronic upper states. This includes sequence bands such as $0^0_0 + h'_g(8) - h''_g(8)$ which could appear close to the expected, forbidden origin frequency 0^0_0 if the upper and lower state mode frequencies differed relatively little. (We remark that the room temperature bands at $\lambda > 410$ nm have fwhm of the order of 160–200 cm^{-1}, see later). These and other hot bands should disappear or be severely attenuated at low temperatures.

Good low temperature spectra were obtained over the range 15500–17000 cm^{-1} for *n*-hexane solutions of C_{60} and 14000–25000 cm^{-1} for 3-methylpentane solutions at 77 K. The 77 K absorption spectra, compared with the room temperature spectra, are shown in fig. 5 for the frequency ranges 14000–18000 cm^{-1} and 23000–25000 cm^{-1}. The fwhm of the band peaks at $\lambda > 410$ nm, which are 120–200 cm^{-1} at room temperature, are reduced to 70–120 cm^{-1} in 3-methylpentane and to about 125 cm^{-1} in *n*-hexane, at 77 K.

The band peak frequencies at 300 K are not significantly modified at 77 K, indicating that if the room temperature spectra include sequence bands the latter either contribute little, or correspond to vibronic transitions for which the vibrational frequencies are similar in upper and lower electronic states.

At first glance it is surprising that all of the bands observed at room temperature between 14000 and 18000 cm^{-1} (fig. 3) appear also in the low temperature spectra (fig. 5); some of them are indeed better

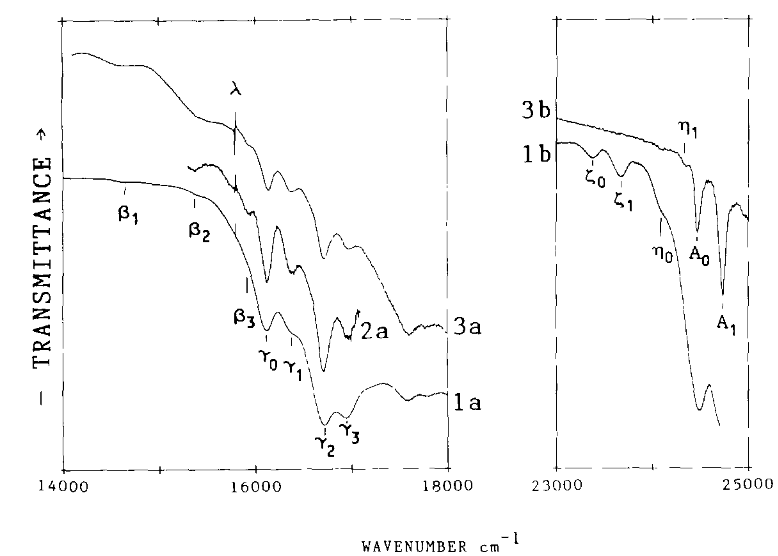

Fig. 5. Room and low temperature spectra of C_{60} solutions measured as transmittance with a Bruker IFS 120 HR spectrometer, $l = 10$ cm, $R = 1\ cm^{-1}$. λ = He-Ne laser line. (1) Room temperature in *n*-hexane: (a) 14000–18000 cm^{-1}, (b) 23000–25000 cm^{-1}. (2) 77 K in polycrystalline *n*-hexane: (a) 15000–17000 cm^{-1}. (3) 77 K in glassy 3-methylpentane: (a) 14000–18000 cm^{-1}, (b) 23000–25000 cm^{-1}.

resolved at low temperature (e.g. γ_1). In addition there are bands which are clearly seen in the low temperature but not in the 300 K spectra. For example, a band whose peak is at 15935 ± 10 cm^{-1} occurs at a frequency which would place it on the flank of the room temperature γ_0 band at 16124 cm^{-1} whose fwhm is about 200 cm^{-1}. The profile of the γ_0 band at 300 K probably englobes a hot band γ_0–$h_g(8)$ at $16124 - 273 = 15851$ cm^{-1}, in addition to γ_0. The hot band level should have a population of about 27% of the vibrationless state at 300 K. The disappearance of the hot band at 77 K, and the smaller bandwidths at low temperatures, allows the 15935 cm^{-1} feature to be seen. Its assignment is discussed in section 4.5.

The two bands ζ_0 and ζ_1 in the 23000–25000 cm^{-1} region (fig. 5) are not seen in the low temperature spectra. They cannot be considered necessarily as hot bands since in the 300 K spectra their low intensities relative to the A_0 band would make them of comparable intensity to the noise in the 77 K 3-methylpentane solution spectrum. We note in the latter that the η_0 band is barely visible above the noise.

We conclude that the strongest hot bands (involving $h_g''(8)$) can remain undetected in the room temperature spectra since they occur unresolved on the flank of the principal peaks.

4.5. The spin-forbidden transitions

The existence of a large number of triplet electronic states is predicted for C_{60} [3,5,6]. The lowest triplet state is calculated to lie at 2.06 eV [5], 2.23 eV [6] or 1.51 eV [7] by, respectively, QCFF/PI, 3D-PPP-CI and INDO/CI methods. From a triplet–triplet energy transfer study the experimental value of the T_1 level energy is reported to lie between 1.43 and 1.82 eV [30].

The first band, β_0, in the $\tilde{\beta}$ system assigned to a triplet$\leftarrow$singlet transition band is at 1.82 eV, i.e. on the upper limit of the experimental T_1 energy range [30]. Our estimate for the oscillator strength of the $\tilde{\beta}$ system is $f(\tilde{\beta}) \lesssim 10^{-4}$. This is of the same order of magnitude as the approximate upper limit value, $f \leqslant 5 \times 10^{-5}$, that we have determined from the lower limit lifetime $\tau \approx 0.3$ ms of the lowest triplet state measured by Wasielewski et al. in EPR experiments [31]. (However, it is not proven that the triplet in the EPR experiment is the same as that determined by the energy transfer experiment.)

If the $\tilde{\beta}$ system corresponds to the lowest $T_1 \leftarrow S_0$ transition, our experimental S_1–T_1 energy gap would be 0.18 eV. Although this is close to the 0.22 eV value calculated by Laszlo and Udvardi (3D-PPP-CI) [6], it differs from that of Negri et al. [5] (0.52 eV; QCCF/PI) and Feng et al. [7] (0.64 eV; INDO/CI).

MO calculations indicate that there should be a number of triplet states in the Y region of singlet states (fig. 4) and that other triplets should be close to the Z group of singlets. Triplet$\leftarrow$singlet transitions should be at least 2–3 orders of magnitude weaker than the singlet–singlet transitions. We therefore expect to detect in absorption only those triplet states that lie below the first excited singlet, i.e. within the S_1–T_1 gap mentioned above.

Further evidence for the assignment of triplet–singlet transitions at energies below $S_1 \leftarrow S_0$ is given by intensity modifications we observed in the oxygen "saturated" solutions. The results will be presented and discussed in detail elsewhere. A brief mention suffices here.

Enhanced T$\leftarrow$S absorption was observed not only in the $\tilde{\beta}$ system region but also between 679 and 695 nm, i.e. to the red of the β_0 band. Thus the lowest triplet state of C_{60} may lie at least as low as 14390 cm^{-1} (1.78 eV) giving an experimental S_1–T_1 gap $\geqslant 0.22$ eV).

We mention that O_2-enhancement of T–S transition intensities also occurs in the 630 nm region and at specific wavelengths between 547 and 615 nm. The enhanced bands at $\lambda \leqslant 615$ nm are considered to correspond to weak T$\leftarrow$S transitions that underly the much stronger forbidden singlet–singlet transition bands. Increased intensity of underlying continuous absorption at $\lambda < 540$ nm is possibly due to C_{60}–O_2 contact charge transfer.

We now discuss the vibronic structure to be expected from triplet–singlet transitions.

4.5.1. Vibronic structure of triplet$\leftarrow$singlet transitions

The lowest T–S transitions, $^3T_{2g}$–^1A_g and $^3T_{1g}$–^1A_g are both spin forbidden and orbitally forbidden. Spin-forbiddenness can be partially lifted by intervention of the spin–orbit coupling operator. This operator must have the symmetry of a rotation which, in the I_h point group, corresponds to the T_{1g} irreducible representation. Excitation of a vibrational mode of suitable symmetry is necessary in order to lift the orbital forbiddenness. The product $\Gamma(SO)\,\Gamma(vib)\,\Gamma(e)$ must contain a symmetry T_{1u} for the T–S transition to be seen. For $\Gamma(e)=T_{1g}$, $\Gamma(vib)=a_u, t_{1u}, t_{2u}, g_u, h_u$ whereas for $\Gamma(e)=T_{2g}$, $\Gamma(vib)=t_{1u}, t_{2u}, g_u, h_u$ (table 3).

The observed false origin bands of the two lowest T_1–S_0 transitions will therefore be associated with excitation of one quantum of appropriate vibrational modes whose $\Gamma(vib)$ symmetries are given above (see also table 3). The lowest frequency modes of these symmetries are expected to have values in the 350–530 cm^{-1} region [5,22]. Since our low temperature spectra did not reveal any T–S hot bands, the true origin of the lowest observed triplet state could be about 0.05 eV below the first observed vibronic transition.

We have assigned the $\tilde{\beta}$ system to the $1\,^3T_{1g}$–$1\,^1A_g$ transition and the lower-lying oxygen induced transition to a weaker $1\,^3T_{2g}$–$1\,^1A_g$ transition, henceforward called $\tilde{\alpha}$ (table 1). Associated with the vibronic "false origins" will be bands where h_g Jahn–Teller active modes and totally symmetric a_g modes are excited in the upper state. The β_1–β_0 interval of 246 cm^{-1} can be assigned to the $h_g(8)$ mode in the $1\,^3T_{1g}$ electronic upper state. The β_2 and β_3 bands correspond to excitation of other h_g modes, e.g. β_2–β_0 = 762 cm^{-1} which is close to $h_g(5)=774$ cm^{-1} of the ground state and β_3–β_0 = 1274 cm^{-1}, close to the ground state $h_g(3)=1250$ cm^{-1} [22]. However, it is also possible that β_3 is the false origin of another triplet–singlet transition, especially since it is close to a region of O_2-enhanced intensity in the 630 nm region.

4.6. Recent relevant work

Since this research was completed and initially written up several works of relevance to our subject have come to our attention and require some comment.

Reber et al. [32] have studied the low temperature (20 K) absorption and luminescence spectra of solid films of C_{60} deposited on a CaF_2 substrate. Nine absorption bands were observed in the 14420–17320 cm^{-1} region; they are in the same general area as our bands α through γ_4. The nine bands are variously displaced in frequency with respect to the "corresponding" bands in our spectra although the relative band intensities are reasonably correlated in the two spectra.

Whetten et al. [33] have further refined an earlier report [11] on the solution absorption of C_{60} and have also examined 77 K absorption of C_{60} in methylcyclohexane–trimethylpentane solutions between 380 and 700 nm. Their low temperature results in the 560–685 nm region are in good agreement with our bands β_0, β_1, γ_0, γ_1, γ_2, γ_3, γ_5, and γ_6. Other bands we observed in this region are less clearly seen (β_2) or are not reported (β_3, γ_4) by Whetten et al. Agreement is also good for most of the bands in the 377–428 nm region.

Haufler et al. [34] have studied the electronic spectra of supersonically cooled C_{60} by the R2PI technique in the 375–415 nm and the 595–640 nm regions; they also obtained absorption spectra in the 380–410 nm region of C_{60} in methylcyclohexane–isopentane solid solutions at 77 K. The complex R2PI spectrum in the 595–640 nm region is still under investigation and analysis; the spectrum between 375 and 415 nm resembles closely, but shifted 400 cm^{-1} to the blue, the 77 K solid solution spectrum in the common spectral region.

Our 77 K absorption spectra extend, to the blue, only up to 25800 cm^{-1} (387.5 nm). In the common region the results are in very reasonable agreement with those of both Whetten et al. [33] and Haufler et al. [34]. At frequencies above 25800 cm^{-1} (i.e. between 380 and 387.5 nm), comparison of our 300 K data with the two 77 K absorption spectra show some small discrepancies (up to 60 cm^{-1}) in the frequencies reported among the three spectra. Inspection of the three spectra indicates that the diminished quality of the low temperature spectra in this region could be responsible, in part, for these discrepancies.

The luminescence spectra of C_{60} on CaF_2 observed by Reber et al. [32] at 20 K show an emission onset at about 14500 cm^{-1} with an initial peak at 14150 cm^{-1}. The spectral distribution was observed to be modified in different regions of the same sample; the resulting spectra were red shifted by up to 500 cm^{-1}. C_{60} emission spectra were also obtained in frozen organic glasses at 20 K. They were similar to those observed for the films. However, Wasielewski et al. [31] report that they observed no phosphorescence for C_{60} solutions in degassed toluene excited by 515 nm radiation at 5 K even though an EPR spectrum of this solution was obtained. (This failure to observe phosphorescence is possibly related to an unusually strong interaction reported to exist between C_{60} and toluene [35].)

The first luminescence peak seen by Reber et al. which occurs at 14150 cm^{-1} or at 13700 cm^{-1} [32] should correspond to false origins of triplet→singlet transitions. Following our O_2-induced results, one possible assignment is that the 14150 cm^{-1} peak corresponds to the false origin of a $1\ ^3T_{2g} \rightarrow 1\ ^1A_g$ transition and that the 13700 cm^{-1} peak is due to an O_2-perturbed $1\ ^3T_{1g} \rightarrow 1\ ^1A_g$ transition.

Finally we discuss briefly the work of Gasyna et al. [29]. Their absorption spectra of C_{60} in argon matrix at 4 K is very similar to our own absorption spectra in *n*-hexane at 300 K and at 77 K but blue shifted with respect to these latter by about 150–200 cm^{-1} for the weak transitions and by about 250–700 cm^{-1} for the strong transitions. We have already mentioned that their magnetic circular dichroism spectra of C_{60} in Ar matrix have provided key evidence for assigning the lowest energy forbidden singlet–singlet transition to $1\ ^1T_{1g}$–$1\ ^1A_g$. One substantial disagreement concerns the assignment of the first allowed transition $1\ ^1T_{1u}$–$1\ ^1A_g$ which Gasyna et al. place at 326 nm (our 328.4 nm) whereas we assigned this transition to the 408 nm band region. Our assignment is based on a comparison with the results on transition energies and oscillator strengths (table 2) obtained by very extensive (900 configurations) CNDO/S calculations by Braga et al. [8], whereas the assignment of Gasyna et al. is based on the QCFF/PI calculations of Negri et al. [5] who used only 196 configurations and whose calculated oscillator strengths are consequently much less reliable (see the discussion in section 4.2). Gasyna et al. tentatively assign the 400 nm region bands to the $1\ ^1T_{2u}$–$1\ ^1A_g$ transition; we point out that the MCD data on which this is based is not incompatible with an assignment to $1\ ^1T_{1u}$–$1\ ^1A_g$.

In the present study we have examined the electronic spectra of C_{60} in more detailed and consistent fashion than in previous work. Thus it is not surprising that our extensive vibronic analysis often differs in important aspects and in detail from the fragmentary analyses to be found in the work discussed in this section.

5. Conclusions

Interest in the electronic spectroscopy of the fullerenes is intense, in part because of the possibility that they can be observed in the interstellar medium and in carbon star regions [36–38]. The present study investigates solution spectra of the fullerene C_{60} at room temperature and at 77 K. The observed spectra exhibit a vast range of band widths and intensities in the wavelength range 195–700 nm. Analysis of the spectra was carried out using the results of available theoretical calculations on C_{60} electronic energy levels and transition intensities. This analysis was pursued within the theoretical frameworks appropriate to three types of transitions: (1) orbitally allowed singlet←singlet transitions exhibiting dynamic Jahn–Teller effects; (2) orbitally forbidden singlet←singlet transitions that occur through Herzberg–Teller vibronic interactions and which undergo Jahn–Teller distortions; (3) orbitally and spin forbidden triplet–singlet transitions requiring spin–orbit and vibronic interactions for their appearance and subject to Jahn–Teller effects.

Low temperature studies and measurements on oxygen-saturated solutions of C_{60} aided the analysis and its refinement. Other types of electronic spectra of C_{60} are certainly required, in particular high resolution gas phase spectra over an extensive energy range, in order to further clarify and understand the electronic spectroscopy of this fullerene. The present study and its results provides a framework for analyses of such spectra.

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References

[1] R.C. Haddon, L.E. Brus and K. Raghavachari, Chem. Phys. Letters 125 (1986) 459.
[2] M. Ozaki and A. Takahashi, Chem. Phys. Letters 127 (1986) 242.
[3] I. Laszlo and L. Udvardi, Chem. Phys. Letters 136 (1987) 418.
[4] S. Larsson, A. Volosov and A. Rosén, Chem. Phys. Letters 137 (1987) 501.
[5] F. Negri, G. Orlandi and F. Zerbetto, Chem. Phys. Letters 144 (1988) 31.
[6] I. Laszlo and L. Udvardi, J. Mol. Struct. 183 (1989) 271.
[7] J. Feng, J. Li, Z. Wang and M.C. Zerner, Intern. J. Quantum Chem. 37 (1990) 599;
J. Feng, J.Li, Z. Li and M.C. Zerner, Intern. J. Quantum Chem. 39 (1991) 331.
[8] M. Braga, S. Larsson, A. Rosén and A. Volosov, Astron. Astrophys. 245 (1991) 232.
[9] W. Krätschmer, K. Fostiropoulos and D.R. Huffman, Chem. Phys. Letters 170 (1990) 167.
[10] W. Krätschmer, L.F. Lamb, K. Fostiropoulos and D.R. Huffman, Nature 347 (1990) 354.
[11] H. Ajie, M.M. Alvarez, S.J. Anz, R.D. Beck, F. Diederich, K. Fostiropoulos, D.R. Huffman, W. Krätschmer, Y. Rubin, K.E. Schriver, D. Sensharma and R.L. Whetten, J. Phys. Chem. 94 (1990) 8630.
[12] J.P. Hare, H.W. Kroto and R. Taylor, Chem. Phys. Letters 177 (1991) 394.
[13] P.M. Allemand, A. Koch, F. Wudl, Y. Rubin, F. Diederich, M.M. Alvarez, S.J. Anz and R.L. Whetten, J. Am. Chem. Soc. 113 (1991) 1050.
[14] R. Taylor, J.P. Hare, A.K. Abdul-Sada and H.W. Kroto, J. Chem. Soc. Chem. Commun. (1990) 1423.
[15] G.E. Scuseria, Chem. Phys. Letters 176 (1991) 423.
[16] G. Herzberg, Electronic Spectra and Electronic Structure of Polyatomic Molecules (Van Nostrand, Princeton, NJ, 1966) p. 50 and p. 573.
[17] R.S. Mulliken and W.B. Person, Molecular Complexes (Wiley/Interscience, New York, 1969).
[18] J.B. Birks, Photophysics of Aromatic Molecules (Wiley/Interscience, New York, 1970).
[19] H.C. Longuet-Higgins, Proc. Roy. Soc. A 255 (1960) 63.
[20] R.F. Curl and R.E. Smalley, Science 242 (1988) 1017.
[21] D.S. Bethune, G. Meijer, W.C. Tang and H.J. Rosen, Chem. Phys. Letters 174 (1990) 219.
[22] D.S. Bethune, G. Meijer, W.C. Tang, H.J. Rosen, W.G. Golden, H. Seki, C.A. Brown and M.S. de Vries, Chem. Phys. Letters 179 (1991) 181.
[23] C. Coulombeau and A. Rassat, J. Chim. Phys. 84 (1987) 875.
[24] Z.C. Wu, D.A. Jelski and T.F. George, Chem. Phys. Letters 137 (1987) 291.
[25] R.E. Stanton and M.F. Newton, J. Am. Chem. Soc. 92 (1988) 2141.
[26] D.E. Weeks and W.G. Harter, J. Chem. Phys. 90 (1989) 4744.
[27] Z. Slanina, J.M. Rudzinski, M. Togasi and E. Osawa, J. Mol. Struct. THEOCHEM 202 (1989) 169.
[28] A.R. Bacon, J.M. Hollas and T. Ridley, Can. J. Phys. 62 (1984) 1254.
[29] Z. Gasyna, P.N. Schatz, J.P. Hare, T.J. Dennis, H.W. Kroto, R. Taylor and D.R.M. Walton, Chem. Phys. Letters 183 (1991) 283.
[30] J.W. Arbogast, A.O. Darmanyan, C.S. Foote, Y. Rubin, F.N. Diederich, M.M. Alvarez, S.J. Anz and R.L. Whetten, J. Phys. Chem. 95 (1991) 11.
[31] M.R. Wasielewski, M.P. O'Neil, K.R. Lykke, M.J. Pellin and D.M. Gruen, J. Am. Chem. Soc. 113 (1991) 2774.
[32] C. Reber, L. Yee, J. McKiernan, J.I. Zink, R.S. Williams, W.M. Tong, D.A.A. Ohlberg, R.L. Whetten and F. Diederich, J. Am. Chem. Soc. 95 (1991) 2127.
[33] R.L. Whetten, M.M. Alvarez, S.J. Anz, K.E. Schriver, R.D. Beck, F.N. Diederich, Y. Rubin, R. Ettl, C.S. Foote, A.P. Darmanyan and J.W. Arbogast, Mater. Res. Soc. Symp. Proc. (1991), in press.
[34] R.E. Haufler, Y. Chai, L.P.F. Chibante, M.R. Fraelich, R.B. Weisman, R.F. Curl and R.E. Smalley, J. Chem. Phys. 95 (1991) 2197.
[35] J. Milliken, T.M. Keller, A.P. Baronowski, S.W. McElvaney, J.H. Callahan and H.H. Nelson, Chem. Mater. (1991), submitted.
[36] H.W. Kroto, in: Polycyclic Aromatic Hydrocarbons and Astrophysics, eds. A. Léger, L. d'Hendecourt and N. Boccara (Reidel, Dordrecht, 1987) p. 197.
[37] A. Léger, L. d'Hendecourt, L. Verstraete and W. Schmidt, Astron. Astrophys. 203 (1988) 145.
[38] H. Kroto, Science 242 (1988) 1139.
[39] S. Larsson, private communication.